HAONAN CHEN

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EDUCATION

Ph.D. in Electrical and Computer Engineering
University of Illinois Urbana-Champaign 08/2020 - Present

B.Sc. in Electrical Engineering with Highest Honors
University of Illinois Urbana-Champaign 09/2016 - 05/2020
GPA: 3.98/4.0

B.Eng. in Electrical Engineering and Automation
Zhejiang University 09/2016 - 06/2020
GPA: 3.99/4.0 (Rank: 1/30)

PUBLICATIONS AND PREPRINTS

[**Under Review**] H. Chen, C. Zhu, Y. Li, K. Driggs-Campbell. "Tool-as-Interface: Learning Robot Tool Use from Human Play through Imitation Learning." *Under Review,* 2025.
Best Presentation Award at CSLSC 2025 [Link]

[**ICRA'25**] H. Chen, J. Xu*, L. Sheng*, T. Ji, S. Liu, Y. Li, K. Driggs-Campbell. "Learning Coordinated Bimanual Manipulation Policies using State Diffusion and Inverse Dynamics Models." *IEEE International Conference on Robotics and Automation,* 2025.

[**JATS'24**] A. Hasan, N. Chakraborty, H. Chen, J.-H. Cho, C. Wu, K. Driggs-Campbell. "Cooperative Advisory Residual Policies for Congestion Mitigation." *ACM Journal on Autonomous Transportation Systems*, 2024. [pdf]

[**Under Review**] A. Hasan, N. Chakraborty, H. Chen, J.-H. Cho, C. Wu, K. Driggs-Campbell. "Lessons in Cooperation: A Qualitative Analysis of Driver Sentiments towards real-time Advisory Systems through a Focus Group User Study." Under Review, 2024. [pdf]

[**CoRL'23**] H. Chen, Y. Niu*, K. Hong*, S. Liu, Y. Wang, Y. Li, K. Driggs-Campbell. "Predicting Object Interactions with Behavior Primitives: An Application in Stowing Tasks." *2023 Conference on Robot Learning.* [pdf] [Project & Video]
Finalist - Best Paper/Best Student Paper Awards

[**CASE'23**] Z. Huang, Y. Mun, H. Chen, Y. Xie, Y. Niu, X. Li, N. Zhono, H. You, D.L. McPherson, K. Driggs-Campbell. "Towards Safety of Multi-Level Human-Robot Interaction in Industrial Tasks." *IEEE Conference on Automation Science and Engineering (CASE), Special Session on Cyberphysical Manufacturing Networks,* 2023.

[**CASE'23**] H. Chen*, Y. Mun*, Z. Huang, Y. Niu, Y. Xie, D.L. McPherson, K. Driggs-Campbell. "Learning Task Skills and Goals Simultaneously from Physical Interaction." *IEEE Conference on Automation Science and Engineering (CASE), Special Session on Cyberphysical Manufacturing Networks,* 2023.

[**ITSC'23**] A. Hasan, N. Chakraborty*, H. Chen*, J-H. Cho, C. Wu, K. Driggs-Campbell. "PeRP: Personalized Residual Policies For Congestion Mitigation Through Co-operative Advisory Systems." *IEEE International Conference on Intelligent Transportation Systems,* 2023. [pdf] [Website]
Best Presentation Award at CSLSC 2024 [Link]

[**ITSC'22**] H. Chen, T. Ji, S. Liu, K. Driggs-Champbell. "Combining Model-Based Controllers and Generative Adversarial Imitation Learning for Traffic Simulation." *IEEE International Conference on Intelligent Transportation Systems (ITSC 2022).* [pdf]

[**ICRA'22**] S. Liu, P. Chang, H. Chen, N. Chakraborty, K. Driggs-Champbell. "Learning to Navigate Intersections with Unsupervised Driver Trait Inference." *IEEE International Conference on Robotics and Automation,* 2022. [pdf] [Website] [Video]

[**IROS'20**] P. Chang, S. Liu, H. Chen, K. Driggs-Campbell. "Robot Sound Interpretation: Combining Sight and Sound in Learning-Based Control." *IEEE/RSJ International Conference on Intelligent Robots and Systems,* 2020. [pdf] [Video]

TEACHING & ADVISING

Undergraduate Teaching Assistant

- ZJU-UIUC Institute Math 241 – Calculus III — Fall 2018
- ZJU-UIUC Institute ECE 310 – Digital Signal Processing — Fall 2019
- ZJU-UIUC Institute ECE 311 – Digital Signal Processing Lab — Fall 2019

AWARDS AND HONORS

Conference Travel Award, Graduate College, University of Illinois Urbana-Champaign — 2023
Finalist - Best Paper/Best Student Paper Awards, Conference on Robot Learning — 2023
Bronze Tablet Award, University of Illinois Urbana-Champaign — 2020
Excellent Graduate, Zhejiang University — 2020
Student Speaker at Graduation, ZJU-UIUC Institute — 2020
Provincial Government Scholarship, Zhejiang Province, China (Top 3%) — *2019*
Dean's List, University of Illinois Urbana-Champaign — 2019
National Scholarship, China (Top 1.8%) — 2018
Excellent Undergraduate Scholarship, Zhejiang University (Top 3%) — 2018-2019
Academic Excellence Scholarship Award, ZJU-UIUC Institute — 2017-2019
Top Ten Social Practice, Zhejiang University (Top 1.5%) — 2017

SERVICE AND LEADERSHIP

- **Robotics Seminar @ Illinois**
  Organizer — August 2023 - Present
- **Coordinated Science Laboratory Student Conference**
  Organizer — 2024

PROFESSIONAL SERVICE

- **Session Chair:**
  - CASE - IEEE Conference on Automation Science and Engineering, 2023
- **Reviewer:**
  - CoRL - Conference on Robot Learning
  - ICRA - IEEE International Conference on Robotics and Automation
  - ITSC - IEEE International Conference on Intelligent Transportation Systems
  - CASE - IEEE Conference on Automation Science and Engineering

OUTREACH AND EXHIBITIONS

Sweet Gestures: Interactive Cupcake Sprinkling Robot 2024

- Designed and showcased an interactive robotic exhibit at the **Engineering Open House 2024** in Urbana-Champaign, which drew over 20,000 attendees, including kids of all ages from across Illinois, and at the **Society of Women Engineers 2024 (WE24)** Conference in Chicago, attended by more than 11,000 participants.

EXTRA-CURRICULAR SERVICE

Founding Team Leader, Robotic Team for RoboMaster Challenge, ZJU-UIUC Institute 10/2017-06/2018